

Problem 11.15

You are given that

$$\delta_t = \frac{t}{100}.$$

Calculate the value at the end of year 10 of an amount that accumulates to \$1,000 at the end of 15 years.

Solution

Let $V(10)$ denote the value at time 10. To obtain the value at time 10 of \$1,000 payable at time 15, discount the amount using the force of interest from time 10 to time 15:

$$V(10) = 1000 \exp \left(- \int_{10}^{15} \delta_t dt \right).$$

Substitute $\delta_t = t/100$:

$$V(10) = 1000 \exp \left(- \int_{10}^{15} \frac{t}{100} dt \right).$$

Evaluate the integral:

$$\int_{10}^{15} \frac{t}{100} dt = \frac{t^2}{200} \Big|_{10}^{15} = \frac{15^2 - 10^2}{200}.$$

Therefore,

$$\int_{10}^{15} \frac{t}{100} dt = \frac{225 - 100}{200} = \frac{125}{200} = \frac{5}{8}.$$

Thus,

$$V(10) = 1000e^{-5/8} \approx 535.2614285.$$

Hence, the value at the end of year 10 is

$$\boxed{V(10) \approx \$535.26}.$$